RESTRICTION ESTIMATES USING MULTILINEAR KAKEYA

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Abstract. We develop a streamlined framework for proving restriction estimates, which yields stronger results than those obtained by polynomial partitioning methods in higher dimensions.

1. Introduction

Let $n \geqslant 5$, and let B^{n-1} be the unit ball in $\mathbb{R}^{n-1}$. Define

$$Ef(x', x_n) = \int_{B^{n-1}} e^{i(x'\cdot\xi + x_n|\xi|^2)} f(\xi)\, d\xi.$$

The restriction problem asks for the range of p for which

$$\|Ef\|_{L^p(\mathbb{R}^n)} \leqslant C_{n,p}\|f\|_{L^p(B^{n-1})}. \tag{1.1}$$

The conjectured range is $p > 2n/(n-1)$. Polynomial partitioning has led to a sequence of advances in this problem; see, for example, [5, 7, 4]. Wang–Wu [9] introduced a different approach based on two-ends incidence geometry and refined decoupling. We combine this approach with the broad–narrow method [1, 5], inserting Zahl's multilinear Kakeya estimate [10] directly into the broad argument. This avoids the loss that can occur when the multilinear estimate is first converted into a linear Kakeya maximal estimate.

Fix

$$2 \leqslant k \leqslant \left\lfloor \frac{n+2}{2} \right\rfloor, \qquad d = d_{n,k} := \frac{n^2 + k^2 + n - k}{2n}, \tag{1.2}$$

and put $D = (4n-5)(d-1) + (n-1)(n-3)$. Define

$$p_b(n,k) = 2 + \frac{4(2d+n-5)}{D}, \qquad p_N(n,k) = 2 + \frac{4}{2n-k}.$$

Here p_b is the threshold for the transverse broad estimate, whereas p_N is the threshold required by the narrow induction.

Theorem 1.1. *For $n \geqslant 5$, the estimate (1.1) holds whenever*

$$p > \min_{2\leqslant k\leqslant\lfloor (n+2)/2\rfloor} \max\{p_b(n,k), p_N(n,k)\}.$$

The threshold in Theorem 1.1 is $2 + 2.58631445\ldots/n + O(n^{-2})$ as $n \to \infty$. For $n = 9$, our result yields the best-known range for the restriction estimate, namely $p > \frac{2150}{929} = 2.314316469\ldots$.

The broad–narrow step is a substitute for the higher-dimensional planebrush estimate that would be needed for a direct geometric approach. Such an estimate could instead strengthen the Wang–Wu framework beyond the present method.

1.1. **Notation and conventions.** We write $A \lesssim B$ when $A \leqslant CB$ with a constant depending only on fixed parameters. At a geometric scale δ, the notation $A \lessapprox B$ allows $C_\epsilon \delta^{-\epsilon}$ for every $\epsilon > 0$, together with any explicitly stated $\delta^{-C\epsilon_1}$ loss from a two-ends hypothesis; the analogous convention is used at physical scale r. Fixed powers of dynamic densities or angles are not absorbed in this notation. We keep logarithmic losses explicit when tracking the broad parameter. A term $\mathrm{RapDec}(r)$ is bounded by $C_N r^{-N}$ for every prescribed N. All scales are rounded dyadically, and fixed dilations of cubes and tubes are understood. We use aligned grids, so that the smaller cubes partition the larger ones.

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2. Preliminaries

2.1. **Two-ends shadings.** A unit δ-tube is a fixed constant enlargement of a unit line segment by transverse radius δ. We represent paraboloid tube directions with positive last coordinate. An m-parallel, direction-separated family has at most m tubes in each δ-separated angular class. We allow fixed dilations of the angular classes and use a bounded coloring when separation is needed.

A shading $Y(T)$ is a union of cells of a fixed δ-grid contained in a fixed enlargement of T. Let

$$E_Y = \bigcup_{T\in\mathbb{T}} Y(T), \qquad \mu_Y(x) = \sum_{T\in\mathbb{T}} \mathbb{1}_{Y(T)}(x).$$

The shading is λ-dense if $|Y(T)| \sim \lambda|T|$ for every retained tube. Later, for a refined subfamily, λ may instead denote the explicitly defined *average* density; this does not impose a new two-ends hypothesis on that subfamily.

Definition 2.1 (Two ends)**.** *For $0 < \epsilon_2 < \epsilon_1 < 1$, a shading is (ϵ_1, ϵ_2)-two-ends if*

$$|Y(T) \cap B(x, \delta^{\epsilon_1})| \leqslant C\delta^{\epsilon_2}|Y(T)|$$

for every T and every such ball. Harmless fixed dilations and fixed constants are incorporated by decreasing ϵ_2 slightly. The constants in the geometric estimates may depend on ϵ_2.

An m-parallel direction-separated family satisfies the Wolff axioms with error $O(m)$ used in [3, Lemma 4.1]; in particular, $\delta^{n-1}\#\mathbb{T} \lesssim m$.

2.2. **Geometric tools.** We use the following two geometric tools with their small-loss versions.

Lemma 2.2 (Gao–Wu–Xi)**.** *Let $\mathbb{T}$ be an m-parallel direction-separated tube family with a λ-dense two-ends shading. Then* [3, Lemma 4.1] *gives a good set $E_H \subset E_Y$ satisfying*

$$|E_Y \backslash E_H| \lesssim \delta^{\epsilon_1}|E_Y|, \qquad \mu_Y \lessapprox m\lambda^{-3/4}\delta^{-(n-2)/2} \quad \text{on } E_H.$$

If the shadings are unions of cubes from a common δ-grid, then E_H may be chosen from the same grid by multiplicity truncation; see [9, Proposition 3.2].

The last assertion follows by truncating μ_Y, which is constant on each cube; see [9, Proposition 3.2].

Lemma 2.3 (Direction-separated multilinear Kakeya; Zahl)**.** *Let $2 \leqslant k \leqslant n$ and $d = d_{n,k}$. For a direction-separated family of unit δ-tubes,* [10, Theorem 1.3] *gives*

$$\left\| \sum_{T_1,\dots,T_k \in \mathbb{T}} \mathbb{1}_{T_1} \cdots \mathbb{1}_{T_k} |v(T_1) \wedge \cdots \wedge v(T_k)|^{k/d} \right\|^{1/k}_{L^{d/k(d-1)}} \lesssim_\epsilon \delta^{1-n/d-\epsilon} (\delta^{n-1} \#\mathbb{T})^{\frac{n(d-1)}{(n-1)d}}.$$

Here $v(T)$ is a unit vector in the direction of T, and $|v_1 \wedge \cdots \wedge v_k|$ is the k-dimensional volume spanned by these vectors.

2.3. **Broad norms.** Fix $R \gg 1$ and a dyadic $K \sim R^\kappa$, with $\kappa > 0$ small. Let τ range over a finitely overlapping cover of B^{n-1} by caps of radius $c_{n,k}K^{-1}$, where $c_{n,k} > 0$ is sufficiently small and fixed. We still call these K^{-1}-caps. Choose a smooth partition of unity $\sum_\tau \psi_\tau = 1$ with $0 \leqslant \psi_\tau \leqslant 1$ and $\operatorname{supp} \psi_\tau \subset \tau$, and put $f_\tau = f\psi_\tau$.

For a cube b of side K^2, define

$$\mu_{k,A,p}(Ef; b) = \inf_{V_1,\dots,V_A} \max_{\substack{\tau:\, \measuredangle(v(2\tau),V_a) > K^{-1} \\ \text{for all } 1 \leqslant a \leqslant A}} \int_b |Ef_\tau|^p, \tag{2.1}$$

where the infimum is over $(k-1)$-dimensional linear subspaces of $\mathbb{R}^n$, and $v(2\tau)$ denotes the tube directions associated with 2τ. Set

$$\|Ef\|^p_{\mathrm{BL}^p_{k,A,K}(U)} = \sum_{b \subset U} \mu_{k,A,p}(Ef; b).$$

We use a fixed grid and take larger cubes to be unions of its K^2-cubes. The same definition applies when the functions Ef_τ are replaced by other functions indexed by the same caps.

At an intermediate scale r, wave-packet decompositions enlarge frequency supports by $O(r^{-1/2})$. We keep the original cap indices and choose the scales so that these supports remain in 2τ.

The broad norm satisfies the following triangle inequality with a loss in A; see [5, Lemma 4.1].

Lemma 2.4 (Broad triangle inequality)**.** *Let $p \geqslant 1$, let $f = \sum_{j=1}^J f_j$, and let $A_1, \dots, A_J$ be nonnegative integers with $A_1 + \cdots + A_J \leqslant A$. Then*

$$\|Ef\|_{\mathrm{BL}^p_{k,A,K}(U)} \leqslant \sum_{j=1}^J \|Ef_j\|_{\mathrm{BL}^p_{k,A_j,K}(U)}.$$

We also use the standard comparison with transverse cap contributions; see [5, Section 1]. We consider tuples of caps whose doubles are pairwise separated by at least $c_{n,k}K^{-1}$. Such a tuple $(\tau_1, \dots, \tau_k)$ is called transverse if

$$|v_1 \wedge \cdots \wedge v_k| \gtrsim K^{-(k-1)} \quad \text{for all } v_i \in v(2\tau_i).$$

For such a tuple, write $f_i = f_{\tau_i}$.

Lemma 2.5 (Transverse caps)**.** *Let $A \geqslant 1$, $p \geqslant 1$, and let X be a union of cubes from the K^2-grid. Then*

$$\|Ef\|^p_{\mathrm{BL}^p_{k,A,K}(X)} \lesssim \sum_{\substack{\tau_1,\dots,\tau_k \\ \text{transverse}}} \sum_{b \subset X} \min_{1 \leqslant i \leqslant k} \int_b |Ef_i|^p \lesssim \sum_{\substack{\tau_1,\dots,\tau_k \\ \text{transverse}}} \left(\prod_{i=1}^k \int_X |Ef_i|^p \right)^{1/k}.$$

2.4. **Wave packets, local averages of the L^2-norm, and refined decoupling.** At scale r, decompose each f_τ into wave packets f_T associated with $r^{-1/2}$-caps θ and tubes T of length r and radius $r^{1/2+\epsilon_0}$. The cap indices are retained throughout. We suppress rapidly decreasing errors and keep the losses $r^{O(\epsilon_0)}$ caused by the tube enlargement.

We use the standard almost-orthogonality properties of wave packets; see [5, Section 3]. After a bounded coloring of the frequency caps and spatial lattice, every subfamily $\mathcal{S}$ satisfies

$$\left\| \sum_{T\in\mathcal{S}} f_T \right\|_2^2 \sim \sum_{T\in\mathcal{S}} \|f_T\|_2^2, \qquad \sum_T \|f_T\|_2^2 \lesssim \|f\|_2^2.$$

For $r \geqslant 1$, write

$$M_r(f) := \max_{\theta:\, r^{-1/2}\text{-cap}} \|f\|_{L^2_{\mathrm{avg}}(\theta)}.$$

When cap labels are retained after wave-packet selection, we use the same notation for

$$\|f\|_2^2 = \sum_\tau \|f_\tau\|_2^2, \qquad M_r(f)^2 = \max_{\theta:\, r^{-1/2}\text{-cap}} \frac{1}{|\theta|} \sum_\tau \int_\theta |f_\tau|^2.$$

For the original smooth cap decomposition, these quantities are comparable to their scalar counterparts. All packet subfamilies below retain these labels. Almost orthogonality and frequency localization give, for $1 \leqslant t \leqslant r$,

$$M_t\left(\sum_{T\in\mathcal{S}} f_T \right) \lesssim M_r(f). \tag{2.2}$$

After dyadic pigeonholing, assume

$$\|f_T\|_2 \sim a, \qquad \#\{T : \theta(T) = \theta\} \sim m \tag{2.3}$$

for each occupied cap. Then

$$a^2 m \lesssim r^{-(n-1)/2} M_r(f)^2, \qquad a^2 \#\mathbb{T} \lesssim \|f\|_2^2. \tag{2.4}$$

Further subfamilies have at most $O(m)$ packets in each cap and still satisfy (2.4). The pigeonholing costs at most $(\log r)^{O(1)}$, including the loss in A from Lemma 2.4.

Let X be a union of $r^{1/2+\epsilon_0}$-cubes, and let

$$f_i = \sum_{T\in\mathbb{T}_i} f_T, \qquad N_i = \#\mathbb{T}_i,$$

where $\mathbb{T}_i$ is a fixed packet subfamily. For each $T \in \mathbb{T}_i$, let $Y_i(T)$ be the union of the cubes of X meeting T, and define

$$\sum_{T\in\mathbb{T}_i} |Y_i(T)| = \lambda_i \sum_{T\in\mathbb{T}_i} |T|.$$

We count incidences using one fixed sufficiently large dilation of all packet tubes, so that tubes not meeting a cube contribute only rapidly decreasing tails. Local L^2 orthogonality on each cube, summed over X, yields

$$\int_X |Ef_i|^2 \lesssim r^{O(\epsilon_0)} \lambda_i r \, \|f_i\|_2^2. \tag{2.5}$$

Set $q = 2(n+1)/(n-1)$. If every cube of X meets at most μ_i tubes from $\mathbb{T}_i$, refined decoupling [6, Theorem 4.2], in the form [9, Theorem 4.4], gives

$$\begin{aligned}\int_X |Ef_i|^q &\lesssim_\epsilon r^{\epsilon+O(\epsilon_0)}\mu_i^{2/(n-1)}a^q N_i \\ &\lesssim r^{-1+\epsilon+O(\epsilon_0)}(\mu_i/m)^{2/(n-1)}\|f_i\|_2^2 M_r(f)^{q-2}.\end{aligned} \tag{2.6}$$

Here the packet L^q estimate follows by interpolating the standard L^2 and L^∞ bounds. The second inequality follows from (2.4). The packet L^q-norms may be grouped dyadically in the refined decoupling estimate; the resulting logarithmic loss is absorbed in r^ϵ.

For the geometric estimates, rescaling the enlarged tubes to unit length gives tube width $\delta_* = r^{-1/2+\epsilon_0}$. Grouping directions at this width increases the parallel multiplicity by at most $r^{O(\epsilon_0)}$. We use the same rescaled grid for the shadings and the good sets, and include these factors in the displayed $r^{O(\epsilon_0)}$ losses.

3. A multilinear incidence estimate

Fix n, k as in (1.2) and set $d = d_{n,k}$. We estimate the multiplicity of transverse tube families without assuming comparable populations. The estimate will be combined with the hairbrush bound and refined decoupling in Section 4.

Lemma 3.1. *Let $\mathbb{T}_1, \dots, \mathbb{T}_k$ be angularly separated families of unit δ-tubes, and suppose $|v(T_1) \wedge \cdots \wedge v(T_k)| \geqslant \nu > 0$ for every choice $T_i \in \mathbb{T}_i$. Assume that the union of the families has at most m tubes in each δ-separated direction class, where m is a positive integer. Let E be a union of $O(\delta^{-n})$ grid cells of side δ. Assume the shaded multiplicity in family i is comparable to a positive constant μ_i on every cell of E. Set*

$$N_i = \#\mathbb{T}_i, \qquad N = \sum_i N_i, \qquad Z = \frac{\delta^{n-1}N}{m},$$

$$\zeta = \frac{(N_1 \cdots N_k)^{1/k}}{N}, \qquad \Lambda = (\lambda_1 \cdots \lambda_k)^{1/k}, \qquad \mu = (\mu_1 \cdots \mu_k)^{1/k},$$

where $\mu_i|E| \sim \lambda_i \delta^{n-1} N_i$ defines the average density λ_i. If $\min_i \mu_i/m \geqslant C_{n,k}\log(2/\delta)$, then

$$\frac{\mu}{m} \lesssim_\epsilon \nu^{-1}\delta^{-(n-d)-\epsilon}\Lambda^{-(d-1)}\zeta^{-(d-1)}Z^{(d-1)/(n-1)}. \tag{3.1}$$

Proof. Pad each direction class with dummy entries to have m slots, and choose one slot uniformly and independently in each class. For each fixed cell and family i, the sampled shaded multiplicity is a sum of independent Bernoulli variables, one for each direction class, of total expectation comparable to μ_i/m. Chernoff's inequality and a union bound over the polynomially many cells show that all these sampled multiplicities are at least $c\mu_i/m$ simultaneously, with probability greater than $3/4$, when the displayed logarithmic threshold is large enough. The expected total number of sampled tubes is N/m; Markov's inequality makes it at most $8N/m$ with probability at least $7/8$. Thus a realization with both properties exists. A bounded direction coloring handles adjacent direction cells when necessary.

The chosen realization is direction-separated, up to the fixed coloring of direction classes. Apply Lemma 2.3 to its complete tubes, summing over the finitely many

color combinations if necessary, using their shaded incidences only for a lower bound on the integrand on E. Raise the result to $d/(d-1)$:

$$\nu^{1/(d-1)}(\mu/m)^{d/(d-1)}|E| \lesssim_\epsilon \delta^{-(n-d)/(d-1)-\epsilon} Z^{n/(n-1)}. \tag{3.2}$$

Taking geometric means of the incidence identities gives

$$\mu|E| \sim \Lambda\zeta\delta^{n-1}N = \Lambda\zeta m Z.$$

Substitute this into (3.2), cancel, and raise to the power $d-1$. This proves (3.1). □

4. Proof of the main theorem

4.1. **The broad two-ends induction.** We prove the broad estimate by a two-ends induction for the broad measures (2.1), combining Lemma 3.1 with the hairbrush bound and refined decoupling in the two-ends case. The outer scale remains fixed while the intermediate radius decreases, and a slowly growing broad parameter pays for the dyadic decompositions; compare [2, Proposition 4.13].

Proposition 4.1. *Fix n, k satisfying (1.2). Given $\eta > 0$, choose $\kappa > 0$ sufficiently small. Then $a_0 > 0$ may be chosen arbitrarily small so that, with $K \sim R^\kappa$, we have*

$$\|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_R)} \lesssim_\eta R^{2\eta}\|f\|_2^2 M_R(f)^{p_b-2}, \qquad A \geqslant R^{a_0}. \tag{4.1}$$

All constants are independent of R and A.

Proof. **Parameter choices.** Choose $\theta > 0$ with $n\theta < \eta/4$, then $\alpha > 0$ so small that the factors $r^{O(\alpha)}$ below are at most $r^{\eta/4}$. Choose $b_0 \ll \alpha\eta$ and $\epsilon_0 \ll b_0$, and take the arbitrary losses in Lemma 2.3 and refined decoupling sufficiently small. Choose κ so that $K^{O(1)} \leqslant R^{\eta/4}$ and $4\kappa < \theta$. The latter condition ensures that the K^2-grid is finer than the $r^{1/2+\epsilon_0}$-grid at every nonterminal scale. Finally choose $a_0 > 0$ as small as needed. All parameters are fixed independently of R.

For $r \in [R^\theta, R]$ we prove

$$\|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_r)} \leqslant C_\eta R^\eta r^\eta \|f\|_2^2 M_r(f)^{p_b-2}, \qquad A \geqslant r^{a_0}. \tag{4.2}$$

We induct from r to $r_1 = r^{1-\alpha}$, stopping when $r \leqslant R^\theta$ and keeping R and K fixed. For every requested radius we start a descending sequence there; its length is bounded in terms of the fixed parameters, independently of R.

Base case. For $r \lesssim R^\theta$, the trivial bound is $Cr^n\|f\|_2^{p_b}$. Since $\|f\|_2 \lesssim M_r(f)$ and $r^n \leqslant CR^{\eta/4}$, this is absorbed by (4.2).

Pigeonholing. First use the bounded packet colorings and the dyadic selections in (2.3). By the broad triangle inequality, these cost $(\log r)^{O(1)}$ in the norm and in A; we include these factors in the choices and bounds below. We retain the notation f for the selected family and A for its broad parameter, now satisfying $A \gtrsim r^{a_0}(\log r)^{-C}$. Its L^2-norm and local averages are controlled by those of the original family. The packets now have comparable L^2-norms, and the occupied fine caps have comparable packet counts.

For each $r^{1/2+\epsilon_0}$-cube Q, write

$$\mathcal{B}(Q) := \|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(Q)}.$$

By dyadic pigeonholing, choose a collection $\mathcal{Q}$ of such cubes and a number $H > 0$ so that, with

$$X = \bigcup_{Q\in\mathcal{Q}} Q,$$

we have

$$\mathcal{B}(Q) \sim H \quad (Q \in \mathcal{Q}), \qquad \sum_{Q \in \mathcal{Q}} \mathcal{B}(Q) \gtrsim (\log r)^{-C} \|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_r)}.$$

We may assume $H \gtrsim r^{-C_1}\|f\|_2^{p_b}$ for a sufficiently large fixed C_1, since otherwise the desired estimate is immediate.

Partition the x_n axis into intervals J of length $r_1 = r^{1-\alpha}$, aligned with the grid, so that each $Q \in \mathcal{Q}$ lies in one x_n-interval $J(Q)$. For each tube T, define

$$Y_J(T) := \bigcup_{\substack{Q \in \mathcal{Q}:\, J(Q)=J \\ Q \cap T \neq \varnothing}} Q,$$

with fixed tube dilations understood. For dyadic $\lambda, \beta > 0$, set

$$\mathcal{J}_\lambda(T) := \{J : \lambda|T| \leqslant |Y_J(T)| < 2\lambda|T|\}, \qquad \mathbb{T}_{\lambda,\beta} := \{T : \beta \leqslant \#\mathcal{J}_\lambda(T) < 2\beta\}.$$

There are $(\log r)^{O(1)}$ possible pairs (λ, β), and $1 \leqslant \beta \lesssim r^\alpha$.

Choose

$$A' = \left\lfloor \frac{A}{(\log r)^{C_0}} \right\rfloor$$

with C_0 sufficiently large to include all preceding divisions of the broad parameter. For $Q \in \mathcal{Q}$, let $\mathcal{B}_{\lambda,\beta}(Q)$ denote the p_bth-power broad mass, with parameter A', of

$$\sum_{\substack{T \in \mathbb{T}_{\lambda,\beta} \\ J(Q) \in \mathcal{J}_\lambda(T)}} Ef_T$$

on Q, retaining the original cap indices. Summing the packet sums over (λ, β) recovers the original sum on Q, up to rapidly decreasing tails: for each tube meeting Q, the interval $J(Q)$ determines a unique density class and hence a unique count class. The broad triangle inequality therefore gives

$$\mathcal{B}(Q) \lesssim (\log r)^C \sum_{\lambda,\beta} \mathcal{B}_{\lambda,\beta}(Q).$$

Hence, for every $Q \in \mathcal{Q}$, we can choose $\lambda(Q), \beta(Q)$ such that

$$\mathcal{B}_{\lambda(Q),\beta(Q)}(Q) \gtrsim (\log r)^{-C} H.$$

Choose a dyadic number $h(Q)$ satisfying

$$h(Q) \leqslant \mathcal{B}_{\lambda(Q),\beta(Q)}(Q) < 2h(Q).$$

Almost orthogonality and the trivial L^∞ estimate give

$$\mathcal{B}_{\lambda,\beta}(Q) \lesssim r^{C_1}\|f\|_2^{p_b}.$$

Together with the lower bound for H, this implies

$$(\log r)^{-C} H \lesssim h(Q) \lesssim r^C H.$$

Thus $h(Q)$ takes only $O(\log r)$ dyadic values.

Pigeonholing the cubes once more, choose a fixed triple (λ, β, h_0) and a subcollection $\mathcal{Q}_2 \subset \mathcal{Q}$ such that

$$\#\mathcal{Q}_2 \gtrsim (\log r)^{-C} \#\mathcal{Q}, \qquad (\lambda(Q), \beta(Q), h(Q)) = (\lambda, \beta, h_0) \quad (Q \in \mathcal{Q}_2).$$

Define

$$X_2 := \bigcup_{Q \in \mathcal{Q}_2} Q.$$

Since the cubes have equal volume,

$$|X_2| \gtrsim (\log r)^{-C}|X|. \tag{4.3}$$

Moreover,

$$\mathcal{B}_{\lambda,\beta}(Q) \sim h_0 \gtrsim (\log r)^{-C} H \quad (Q \in \mathcal{Q}_2),$$

and consequently

$$\sum_{Q\in\mathcal{Q}_2} \mathcal{B}_{\lambda,\beta}(Q) \gtrsim (\log r)^{-C} \|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_r)}.$$

For a union U of the selected cubes, write

$$\mathcal{B}_{\lambda,\beta}(U) := \sum_{Q\subset U} \mathcal{B}_{\lambda,\beta}(Q).$$

Thus X_2 retains a polylogarithmic fraction of both the volume and the original broad mass, with comparable selected broad mass on each cube.

For $T \in \mathbb{T}_{\lambda,\beta}$, define the shading

$$Y(T) := \bigcup_{J\in\mathcal{J}_\lambda(T)} Y_J(T).$$

Then

$$|Y(T)| \sim \lambda_0|T|, \qquad \lambda_0 := \beta\lambda.$$

These shadings are formed using the original X, not X_2. Every retained cube meets an active tube, after the negligible tails have been discarded, so

$$X_2 \subset \bigcup_{T\in\mathbb{T}_{\lambda,\beta}} Y(T) \subset X.$$

We now relabel $\mathbb{T}_{\lambda,\beta}$ as $\mathbb{T}$ and redefine

$$X := \bigcup_{T\in\mathbb{T}} Y(T).$$

The estimate (4.3) remains valid with this new X. On each common x_n-interval, the selected sum is the extension of a fixed packet subfamily. For sufficiently large r, all preceding parameter losses leave

$$A' \gtrsim \frac{r^{a_0}}{(\log r)^{O(1)}} \geqslant r^{(1-\alpha)a_0} = r_1^{a_0}.$$

Case 1: The non-two-ends case. Suppose $\beta \leqslant r^{b_0}$. Partition each common x_n-slab into r_1-cubes and cover these by balls B_j of radius $O(r_1)$. For a cube in the slab corresponding to J, let f_j be the sum of packets active on J whose tubes meet a fixed dilation of that cube. This agrees with the selected sum on the cube, up to the suppressed tails.

Since each wave packet is selected in at most $O(r^{b_0})$ x_n-intervals, almost orthogonality and (2.2) give

$$\sum_j \|f_j\|_2^2 \lesssim r^{b_0+O(\epsilon_0)}\|f\|_2^2, \qquad M_{r_1}(f_j) \lesssim M_r(f).$$

As $A' \geqslant r_1^{a_0}$, the induction hypothesis (4.2) yields

$$\begin{aligned}\mathcal{B}_{\lambda,\beta}(X_2) &\lesssim \sum_j \|Ef_j\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A',K}(B_j)} \lesssim C_\eta R^\eta r_1^\eta \sum_j \|f_j\|_2^2 M_{r_1}(f_j)^{p_b-2} \\ &\lesssim C_\eta R^\eta r^{(1-\alpha)\eta+b_0+O(\epsilon_0)}\|f\|_2^2 M_r(f)^{p_b-2} \ll R^\eta r^\eta \|f\|_2^2 M_r(f)^{p_b-2},\end{aligned}$$

provided $b_0 + O(\epsilon_0) < \alpha\eta$. The preceding logarithmic losses are absorbed by the strict inequality, which closes (4.2) in this case.

Case 2: The two-ends case. Suppose $\beta > r^{b_0}$, and recall that the ancestor shading Y has density $\lambda_0 \sim \beta\lambda$. Each ball of radius r_1 meets only $O(1)$ common x_n-intervals. Since the selected segment volumes are comparable,

$$|Y(T) \cap B(x, r_1)| \lesssim \beta^{-1}|Y(T)| \lesssim r^{-b_0}|Y(T)|.$$

After rescaling to unit length, this is a two-ends condition at width $\delta_* = r^{-1/2+\epsilon_0}$; for example, take $\epsilon_1 = 3\alpha$ and $\epsilon_2 = b_0$. Applying Lemma 2.2 to Y, we obtain a good set $E_H \subset X$, which is a union of cubes from the same grid as X_2. Since the selected broad mass is comparable on these cubes, (4.3) gives

$$\mathcal{B}_{\lambda,\beta}(X_2 \backslash E_H) \lesssim \frac{|X_2 \backslash E_H|}{|X_2|} \mathcal{B}_{\lambda,\beta}(X_2) \lesssim r^{-c\alpha}(\log r)^{O(1)} \mathcal{B}_{\lambda,\beta}(X_2).$$

For sufficiently large r, this is less than half of the original broad mass. We replace X_2 by $X_2 \cap E_H$ and retain the notation.

By Lemma 2.5 and pigeonholing, fix a common x_n-interval J and a quantitatively transverse tuple of caps $(\tau_1, \dots, \tau_k)$. Let $\mathbb{T}_i$ consist of the packets in τ_i that are active on J, and write

$$f_i = \sum_{T \in \mathbb{T}_i} f_T.$$

Pigeonholing the separate positive multiplicities on the $r^{1/2+\epsilon_0}$-cubes, we obtain a region $X_3 \subset X_2 \cap (\mathbb{R}^{n-1} \times J)$ on which each multiplicity is comparable to a constant μ_i. Cubes where some multiplicity vanishes contribute only rapidly decreasing errors. There are $(\log r)^{O(1)}$ joint multiplicity classes; applying pigeonholing to the sum of minima in Lemma 2.5 gives

$$\mathcal{B}_{\lambda,\beta}(X_2) \lesssim K^{O(1)} r^{O(\alpha)} (\log r)^{O(1)} \sum_{b \subset X_3} \min_{1 \leqslant i \leqslant k} \int_b |Ef_i|^{p_b}. \tag{4.4}$$

Only now do we apply Hölder to obtain

$$\mathcal{B}_{\lambda,\beta}(X_2) \lesssim K^{O(1)} r^{O(\alpha)} (\log r)^{O(1)} \left(\prod_{i=1}^{k} \int_{X_3} |Ef_i|^{p_b} \right)^{1/k}. \tag{4.5}$$

For these families, use the notation $\lambda_i, \Lambda, Z, \zeta$ of Lemma 3.1, with shadings $Y_i(T) = Y(T) \cap X_3$ for $T \in \mathbb{T}_i$. Because $\mathbb{T}_i$ is active on the fixed interval J, these are exactly the incidence shadings on X_3 used in Subsection 2.4. Since $Y_i(T) \subset Y(T)$,

$$\lambda_i \lesssim \lambda_0, \qquad \Lambda \lesssim \lambda_0.$$

The inherited hairbrush estimate is

$$\mu_i \lesssim \mu_Y \lesssim_\epsilon m \lambda_0^{-3/4} r^{(n-2)/4+\epsilon+O(\alpha+\epsilon_0)} \quad \text{on } X_3. \tag{4.6}$$

Thus $\lambda_0^{-3/4}$ may be replaced by $C\Lambda^{-3/4}$.

First suppose $\mu_i/m \lesssim r^{C\epsilon_0} \log r$ for some i. Set $t = (p_b - 2)/(q - 2)$. By (2.5), (2.6), and Hölder's inequality,

$$\int_{X_3} |Ef_i|^{p_b} \leqslant \left(\int_{X_3} |Ef_i|^2 \right)^{1-t} \left(\int_{X_3} |Ef_i|^q \right)^t \lesssim_\epsilon r^{1-2t+\epsilon+O(\epsilon_0)} \|f_i\|_2^2 M_r(f)^{p_b-2}.$$

Since

$$1 - 2t = -\frac{(d-1) + (n-1)(n-3)}{D} < 0,$$

and

$$\sum_{b\subset X_3}\min_j\int_b|Ef_j|^{p_b}\leqslant\int_{X_3}|Ef_i|^{p_b},$$

(4.4) and $\|f_i\|_2^2\lesssim\|f\|_2^2$ give (4.2), with the small parameters chosen as above.

We may therefore assume that $\mu_i/m\gtrsim r^{C\epsilon_0}\log r$ for every i, so that Lemma 3.1 applies. Define

$$\mathcal{I}_\ell:=\left(\prod_{i=1}^k\int_{X_3}|Ef_i|^\ell\right)^{1/k},\qquad \ell\in\{2,q,p_b\}.$$

Packet-size uniformization and almost orthogonality give

$$\left(\prod_{i=1}^k\|f_i\|_2^2\right)^{1/k}\lesssim\zeta\|f\|_2^2.$$

Taking geometric means in (2.5), and combining (2.6) with (4.6) and Lemma 3.1, respectively, yields

$$\begin{aligned}
&\mathcal{I}_2\lesssim r^{1+O(\epsilon_0)}\Lambda\zeta\|f\|_2^2,\\
&\mathcal{I}_q\lesssim_\epsilon r^{-\frac{n}{2(n-1)}+\epsilon+O(\alpha+\epsilon_0)}\Lambda^{-\frac{3}{2(n-1)}}\zeta\|f\|_2^2M_r(f)^{q-2},\\
&\mathcal{I}_q\lesssim_\epsilon K^{O(1)}r^{-\frac{d-1}{n-1}+\epsilon+O(\epsilon_0)}\Lambda^{-\frac{2(d-1)}{n-1}}\zeta^{1-\frac{2(d-1)}{n-1}}\|f\|_2^2M_r(f)^{q-2}.
\end{aligned}\tag{4.7}$$

In the last estimate, we have omitted the bounded factor $Z^{2(d-1)/(n-1)^2}$; the $K^{O(1)}$ loss accounts for quantitative transversality.

Choose

$$u=\frac{(2n-3)(d-1)}{D},\qquad v=\frac{2(n-1)(d-1)}{D},\qquad w=\frac{(n-1)(n-3)}{D}.$$

These weights sum to one, satisfy $2u+q(v+w)=p_b$, and cancel the powers of r and Λ in (4.7), apart from the small losses. Moreover,

$$1-\frac{2(d-1)}{n-1}w=\frac{(2n+1)(d-1)+(n-1)(n-3)}{D}>0.$$

Applying the two bounds for $\mathcal{I}_q$ with weights v and w, respectively, we obtain

$$\mathcal{I}_{p_b}\leqslant\mathcal{I}_2^u\mathcal{I}_q^{v+w}\lesssim_\epsilon K^{O(1)}r^{\epsilon+O(\alpha+\epsilon_0)}\|f\|_2^2M_r(f)^{p_b-2},$$

since $\zeta\leqslant1$. Combining this with (4.5) and the preceding pigeonholing gives

$$\|Ef\|^{p_b}_{\mathrm{BL}^{p_b}_{k,A,K}(B_r)}\lesssim_\epsilon K^{O(1)}r^{\epsilon+O(\alpha+\epsilon_0)}(\log r)^{O(1)}\|f\|_2^2M_r(f)^{p_b-2}.$$

The parameter choices absorb the displayed losses into $C_\eta R^\eta r^\eta$. This proves (4.2) and hence (4.1). □

4.2. **From mixed broad estimates to linear restriction.** Fix

$$\max\{p_b,p_N\}<p<\frac{2(k-1)}{k-2},$$

where the upper endpoint is interpreted as infinity when $k=2$. For the canonical cap decomposition, Proposition 4.1, $M_R(f)\lesssim\|f\|_\infty$, and $\|Ef_\tau\|_\infty\lesssim\|f\|_\infty$ give

$$\|Ef\|^p_{\mathrm{BL}^p_{k,A,K}(B_R)}\lesssim_{\eta,p}R^{2\eta}\|f\|_2^2\|f\|_\infty^{p-2},\qquad K\sim R^\kappa,\quad A\geqslant R^{a_0}.$$

We prove the corresponding mixed linear estimate by induction on R. The broad–narrow decomposition and [5, Lemma 9.3] give

$$\int_{B_R} |Ef|^p \lesssim K^{C_{n,p}} \|Ef\|^p_{\mathrm{BL}^p_{k,A,K}(B_R)} + C_{\delta_N} A^{p-1} K^{\delta_N + (k-2)(p/2-1)} \sum_\tau \int_{B_{2R}} |Ef_\tau|^p,$$

up to rapidly decreasing errors. The factor A^{p-1} comes from assigning each narrow cap to one exceptional plane and applying Hölder over the resulting disjoint groups.

Parabolic rescaling and the induction hypothesis yield

$$\int_{B_{2R}} |Ef_\tau|^p \lesssim C_\epsilon (R/K)^\epsilon K^{2n-(n-1)p} \|f_\tau\|_2^2 \|f_\tau\|_\infty^{p-2}.$$

After summing in τ, the narrow term has the relative factor

$$A^{p-1} K^{-\nu_p + \delta_N - \epsilon}, \qquad \nu_p = \frac{2n-k}{2}(p-2) - 2 > 0.$$

Take $A = \lfloor R^{a_0} \rfloor$ and choose

$$2\eta + \kappa C_{n,p} < \frac{\epsilon}{2}, \qquad \delta_N < \frac{\nu_p}{4}, \qquad a_0(p-1) < \frac{\kappa \nu_p}{4}.$$

The narrow factor is then $O(K^{-\nu_p/2})$, while the broad term is $O(R^{\epsilon/2})\|f\|_2^2 \|f\|_\infty^{p-2}$. Since a_0 may be chosen arbitrarily small after fixing κ, the mixed induction closes. Restricted-type real interpolation and standard ϵ-removal [8], with a local exponent strictly below the desired global exponent, complete the proof.

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